INTRODUCTION AND OVERVIEW OF MANUFACTURING

1. What is Manufacturing?
2. Materials in Manufacturing
3. Manufacturing Processes
4. Production Systems
5. Manufacturing Economics
6. Recent Developments in Manufacturing

Manufacturing is Important

- Making things has been an essential human activity since before recorded history
- Today, the term *manufacturing* is used for this activity
- Manufacturing is important to the United States and most other developed and developing nations
  - Technologically
  - Economically

Technological Importance

- Technology - the application of science to provide society and its members with those things that are needed or desired
  - Technology affects our daily lives, directly and indirectly, in many ways
  - Technology provides the products that help our society and its members live better

Technological Importance

- What do these products have in common?
  - They are all manufactured
  - They would not be available to our society if they could not be manufactured
- Manufacturing is the essential factor that makes technology possible

Economic Importance

U.S. Economy

Sector:	%GDP
Agriculture and natural resources	5
Construction and public utilities	5
Manufacturing	12
Service industries*	78
	100

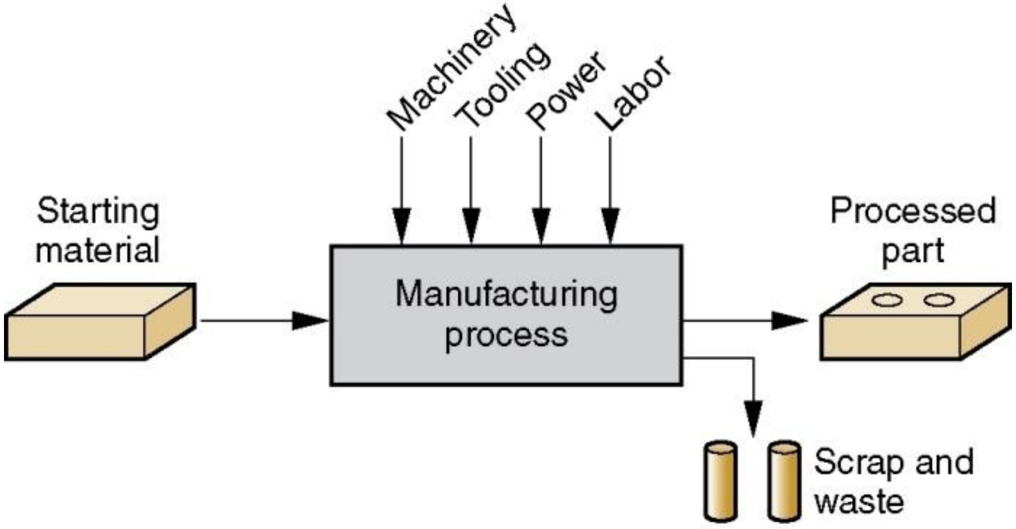
* includes retail, transportation, banking, communication, education, and government

What is Manufacturing?

- The word manufacture is derived from two Latin words *manus* (hand) and *factus* (make); the combination means “made by hand”
  - “Made by hand” described the fabrication methods that were used when the English word “manufacture” was first coined around 1567 A.D.
  - Most modern manufacturing operations are accomplished by mechanized and automated equipment that is supervised by human workers

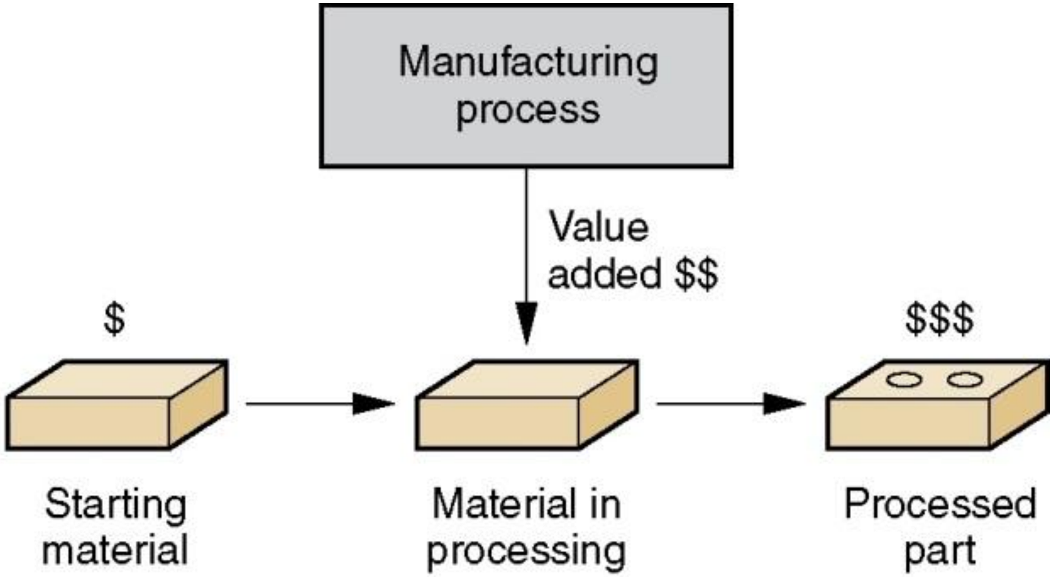
Manufacturing - Technological

- Application of physical and chemical processes to alter the geometry, properties, and/or appearance of a starting material to make parts or products



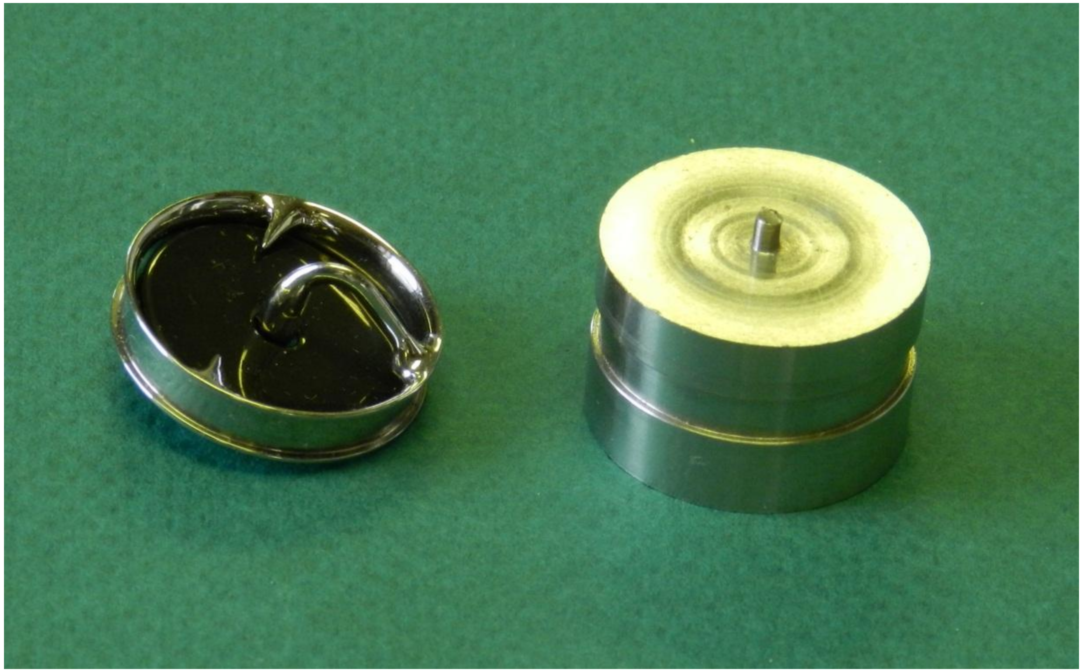
Manufacturing - Economic

- Transformation of materials into items of greater value by one or more processing and/or assembly operations



Manufacturing Example: Artificial Heart Valve

Left: Heart valve   Right: Starting titanium billet



Manufacturing Industries

- Industry consists of enterprises and organizations that produce or supply goods and services
- Industries can be classified as:

1. Primary industries - cultivate and exploit natural resources, e.g., agriculture, mining
2. Secondary industries - take the outputs of primary industries and convert them into consumer and capital goods
3. Tertiary industries - service sector

Specific Industries in Each Category

TABLE 1.2 Specific industries in the primary, secondary, and tertiary categories.

Primary	Secondary		Tertiary (Service)	
Agriculture	Aerospace	Food processing	Banking	Insurance
Forestry	Apparel	Glass, ceramics	Communications	Legal
Fishing	Automotive	Heavy machinery	Education	Real estate
Livestock	Basic metals	Paper	Entertainment	Repair and maintenance
Quarries	Beverages	Petroleum refining	Financial services	Restaurant
Mining	Building materials	Pharmaceuticals	Government	Retail trade
Petroleum	Chemicals	Plastics (shaping)	Health and medical	Tourism
	Computers	Power utilities	Hotel	Transportation
	Construction	Publishing	Information	Wholesale trade
	Consumer appliances	Textiles		
	Electronics	Tire and rubber		
	Equipment	Wood and furniture		
	Fabricated metals			

Manufacturing Industries - continued

- Secondary industries include manufacturing, construction, and electric power generation
- Manufacturing includes several industries whose products are not covered here; e.g., apparel, beverages, chemicals, and food processing
- For our purposes, manufacturing means production of hardware
  - Nuts and bolts, forgings, cars, airplanes, digital computers, plastic parts, and ceramic products

Manufactured Products

- Final products divide into two major classes:
  1. Consumer goods - products purchased directly by consumers
     - Cars, clothes, TVs, tennis rackets
  2. Capital goods - those purchased by companies to produce goods and/or provide services
     - Aircraft, computers, communication equipment, medical apparatus, trucks, machine tools, construction equipment

Production Quantity Q

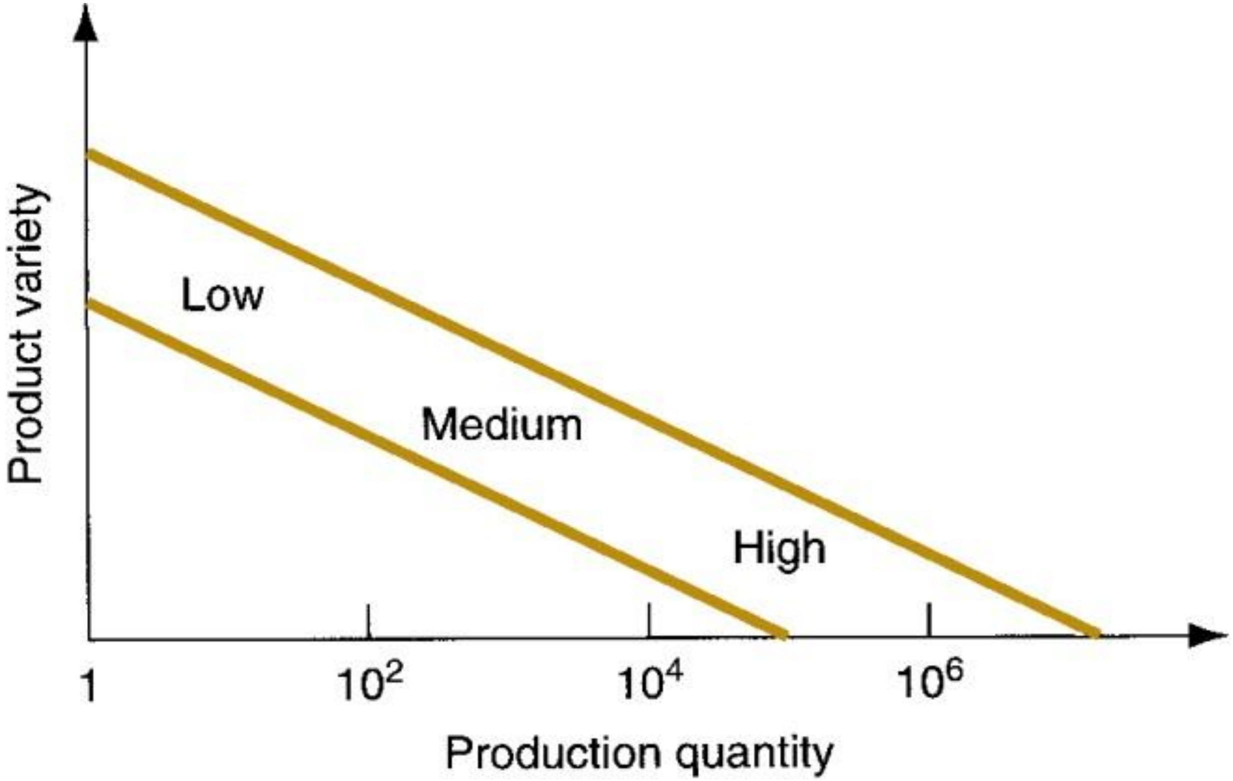
- The quantity of products *Q* made by a factory has an important influence on the way its people, facilities, and procedures are organized
- Annual quantities can be classified into three ranges:

Production range	Annual Quantity Q
Low production	1 to 100 units
Medium production	100 to 10,000 units
High production	10,000 to millions of units

Product Variety *P*

- Product variety *P* refers to different product types or models produced in the plant
- Different products have different features
  - They are intended for different markets
  - Some have more parts than others
- The number of different product types made each year in a factory can be counted
- When the number of product types made in the factory is high, this indicates high product variety

P vs *Q* in Factory Operations



More About Product Variety

- Although *P* is quantitative, it is much less exact than *Q* because details on how much the designs differ is not captured simply by the number of different designs
- *Soft product variety* - small differences between products, e.g., between car models made on the same production line, with many common parts
- *Hard product variety* - products differ substantially, e.g., between a small car and a large truck, with few common parts (if any)

Manufacturing Capability

- A manufacturing plant consists of *processes* and *systems* (and people) to transform a certain limited range of *materials* into products of increased value
- The three building blocks - materials, processes, and systems - are the subject of modern manufacturing
- Manufacturing capability includes:
  1. Technological processing capability
  2. Physical product limitations
  3. Production capacity

1. Technological Processing Capability

- The set of available manufacturing processes in the plant (or company)
- Certain manufacturing processes are suited to certain materials, so by specializing in certain processes, the plant is also specializing in certain materials
- Includes not only the physical processes, but also the expertise of the plant personnel
    - A machine shop cannot roll steel
    - A steel mill cannot build cars

2. Physical Product Limitations

- Given a plant with a certain set of processes, there are size and weight limitations on the parts or products that can be made in the plant
- Product size and weight affect:
  - Production equipment
  - Material handling equipment
- Production, material handling equipment, and plant size must be planned for products that lie within a certain size and weight range

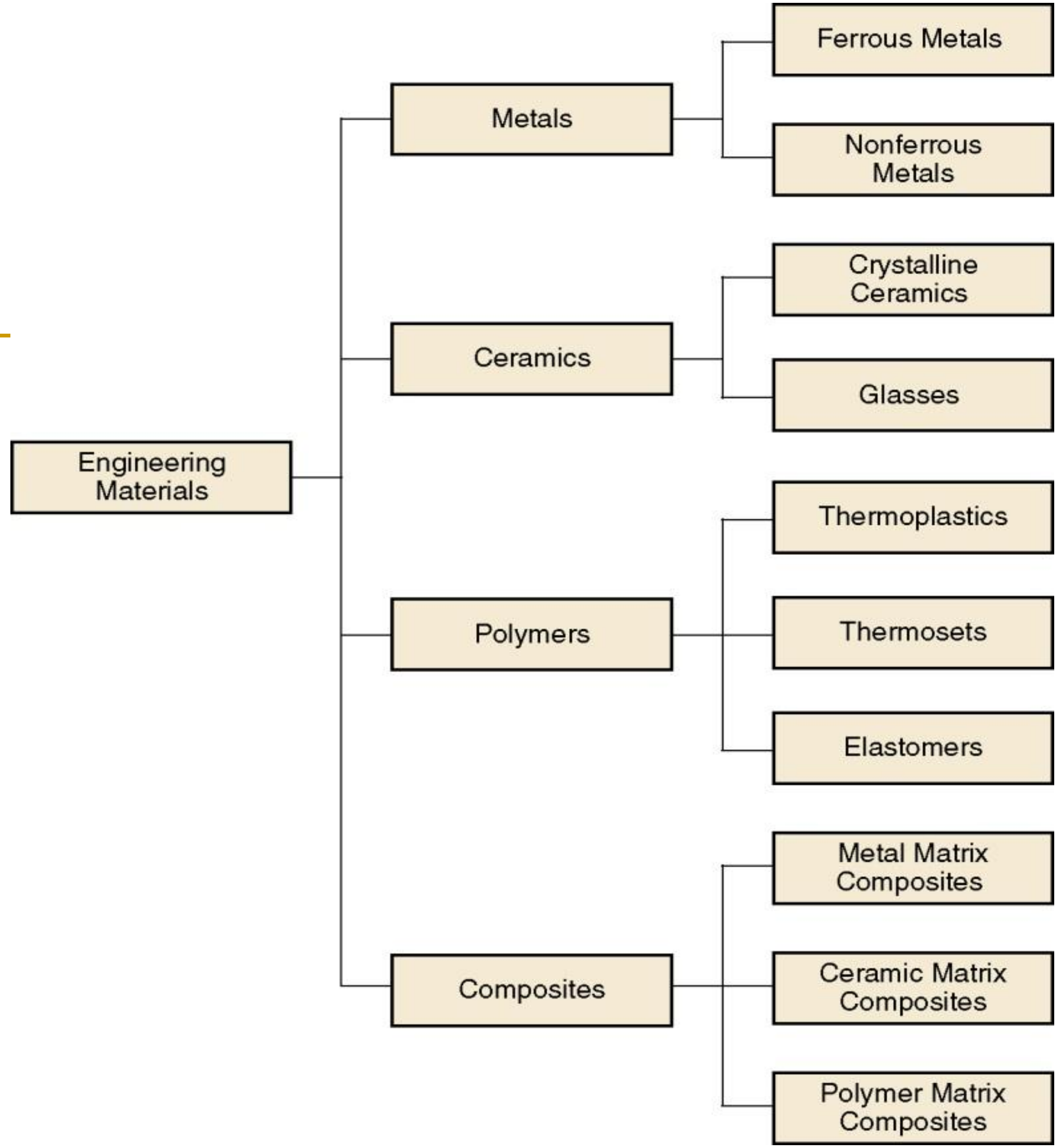
3. Production Capacity

- Defined as the maximum quantity that a plant can produce in a given time period (e.g., month or year) under assumed operating conditions
  - Operating conditions refer to number of shifts per week, hours per shift, direct labor manning levels in the plant, and so on
  - Usually measured in terms of output units, e.g., tons of steel or number of cars produced
  - Also called *plant capacity*

Materials in Manufacturing

- Most engineering materials can be classified into one of three basic categories:
  1. Metals
  2. Ceramics
  3. Polymers
- Their chemistries are different, and their mechanical and physical properties are different
- In addition, there is a fourth category:
  4. Composites

Classification of engineering materials



1. Metals

- Usually *alloys*, which are composed of two or more elements, at least one of which is metallic. Two basic groups:
  1. Ferrous metals - based on iron, comprises about 75% of metal tonnage in the world:
     - Steel and cast iron
  2. Nonferrous metals - all other metallic elements and their alloys:
     - Aluminum, copper, nickel, silver, tin, etc.

Why are alloys not compounds?

Alloys are mixtures **because the elements that make it up are physically together, not chemically combined or joined together as in a compound**. No new chemical properties are shown once the metals are mixed. So, an alloy retains the properties of the constituent elements.

2. Ceramics

- Compounds containing metallic (or semi-metallic) and nonmetallic elements.
  - Typical nonmetallic elements are oxygen, nitrogen, and carbon
  - For processing, ceramics divide into:
    1. Crystalline ceramics – includes traditional ceramics, such as clay, and modern ceramics, such as alumina (Al_2O_3)
    2. Glasses – mostly based on silica (SiO_2)

3. Polymers

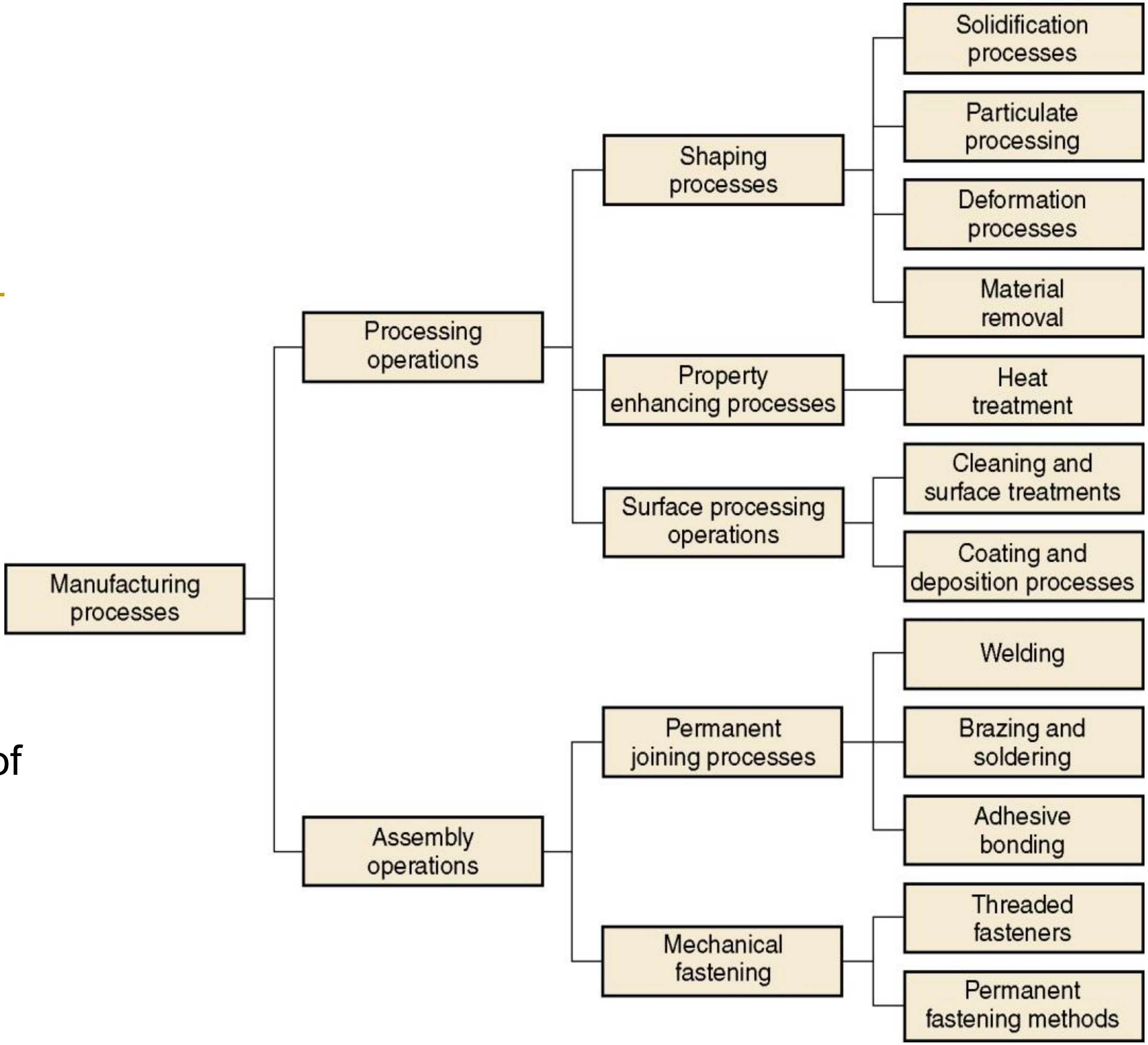
- Compound formed of repeating structural units called *mers*, whose atoms share electrons to form very large molecules. Three categories:
  1. Thermoplastic polymers - can be subjected to multiple heating and cooling cycles without altering molecular structure
  2. Thermosetting polymers - molecules chemically transform into a rigid structure – cannot reheat
  3. Elastomers - shows significant elastic behavior

4. Composites

- Material consisting of two or more phases that are processed separately and then bonded together to achieve properties superior to its constituents
  - *Phase* - homogeneous material, such as grains of identical unit cell structure in a solid metal
  - Usual structure consists of particles or fibers of one phase mixed in a second phase
  - Properties depend on components, physical shapes of components, and the way they are combined to form the final material

Manufacturing Processes: Two Basic Types

1. Processing operations - transform a work material from one state of completion to a more advanced state
   - Operations that change the geometry, properties, or appearance of the starting material
2. Assembly operations - join two or more components to create a new entity



Classification of Manufacturing Processes

Processing Operations

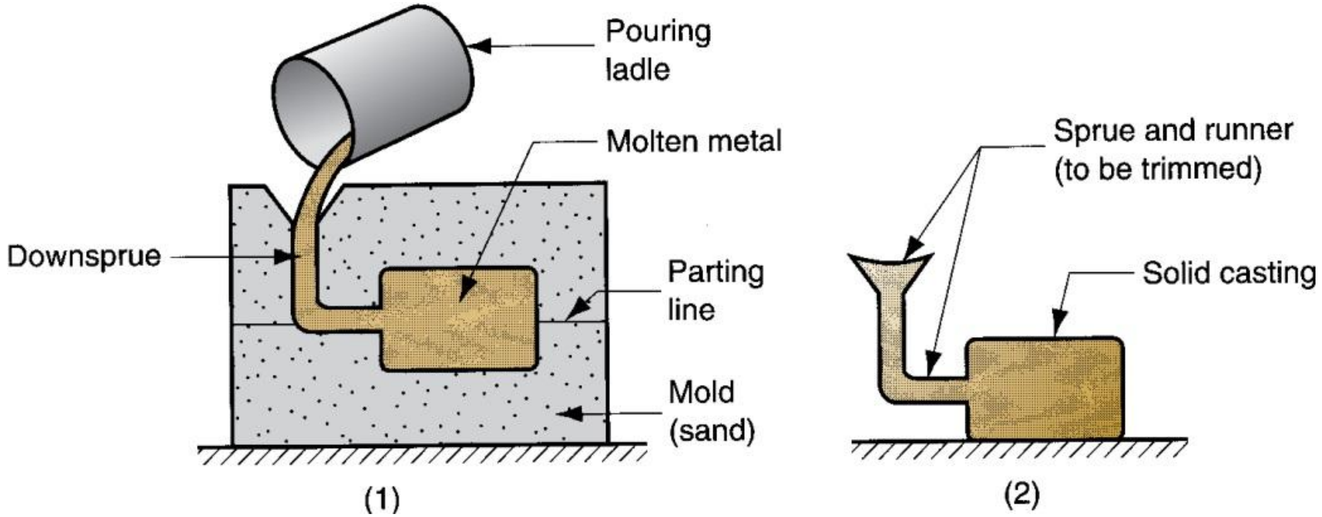
- Alters a material's shape, physical properties, or appearance in order to add value
- Three categories of processing operations:
  1. Shaping operations - alter the geometry of the starting work material
  2. Property-enhancing operations - improve physical properties without changing shape
  3. Surface processing operations - clean, treat, coat, or deposit material on surface of work

Four Categories of Shaping Processes

1. Solidification processes - starting material is a heated liquid or semifluid
2. Particulate processing - starting material consists of powders
3. Deformation processes - starting material is a ductile solid (commonly metal)
4. Material removal processes - starting material is a ductile or brittle solid

Solidification Processes

- Starting material is heated sufficiently to transform it into a liquid or highly plastic state
- (1) Casting process and (2) casting product



Solidification Processes

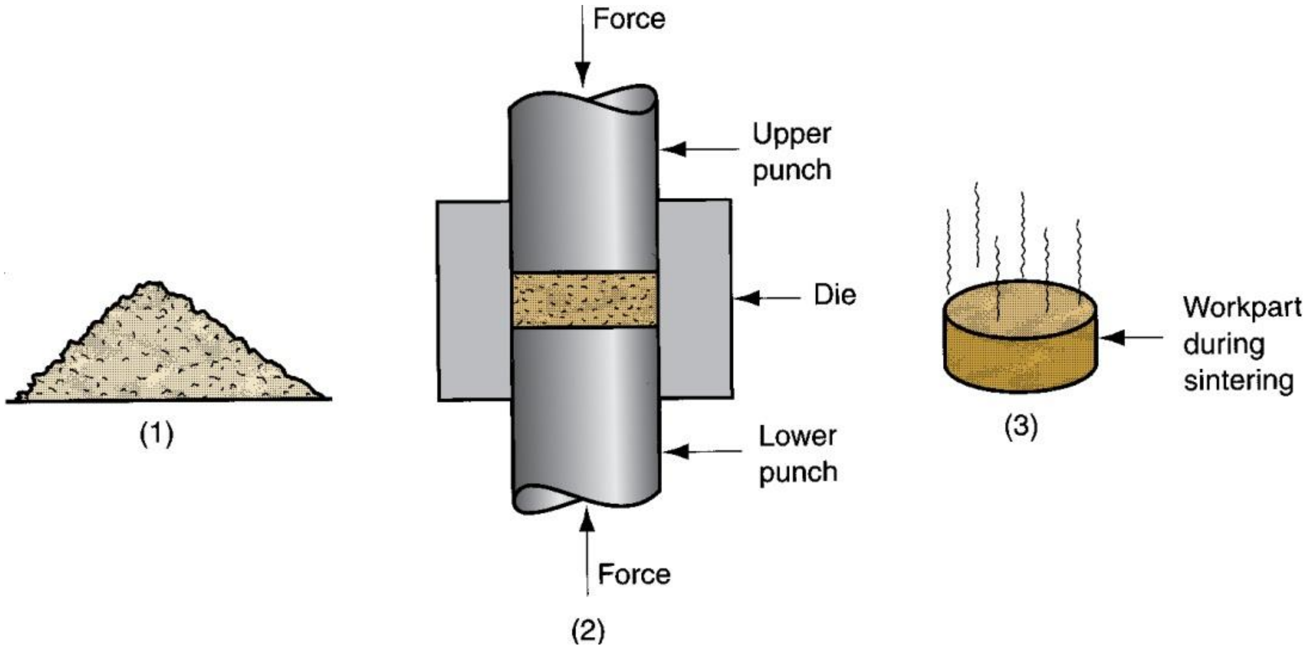
- Casting:
  - Die casting
    - https://www.youtube.com/watch?v=0oibUY8KUQM
    - https://www.youtube.com/watch?v=ThOwfwzess8
  - Sand casting
    - https://www.youtube.com/watch?v=LmjAQGvSrF0

Molding or Moldmaking is the act of creating the cavity / form that carries a negative or reverse impression of an original model. Molds can be made of a rigid material, such as plaster or plastic resin or more commonly, a flexible material such as rubber. The material to use should be chosen considering the material of the model, the material to be used to make castings, and whether there are any undercuts.

Casting is the act of pouring liquid material into the cavity of a mold. After a period of time, this liquid will cure via chemical reaction or cooling. The solidified part is also known as a casting, which is ejected or broken out of the mold to complete the process. Casting materials are usually metals or various cold setting materials that cure after mixing two or more components together; examples are epoxy, concrete, plaster and clay.

Particulate Processing

- (1) Starting materials are metal or ceramic powders, which are (2) pressed and (3) sintered

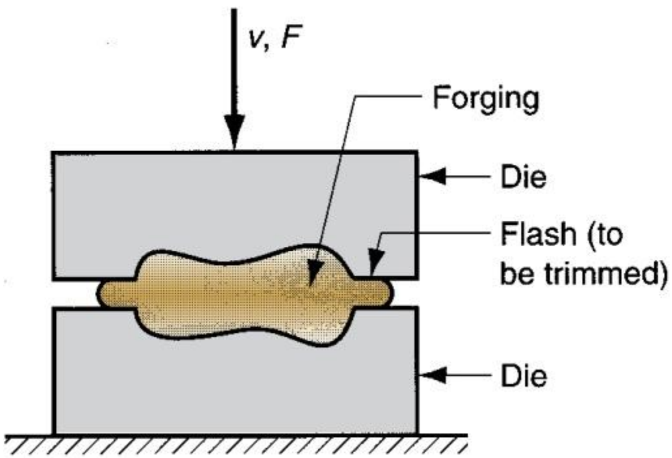


Particulate Processing

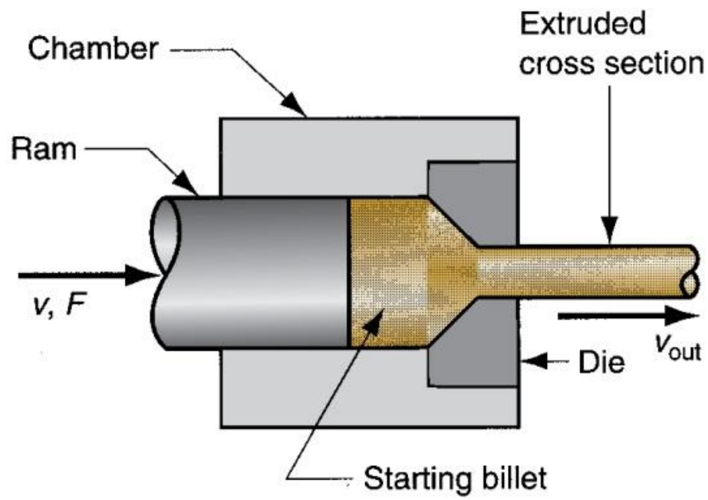
- Powdered metal process:
  - https://www.youtube.com/watch?v=DeM7pRVLqbk
  - https://www.youtube.com/watch?v=azGg68B-GIk

Deformation Processes

- Starting workpart is shaped by application of forces that exceed the yield strength of the material
  - Examples: (a) forging and (b) extrusion



(a)



(b)

Deformation Processes

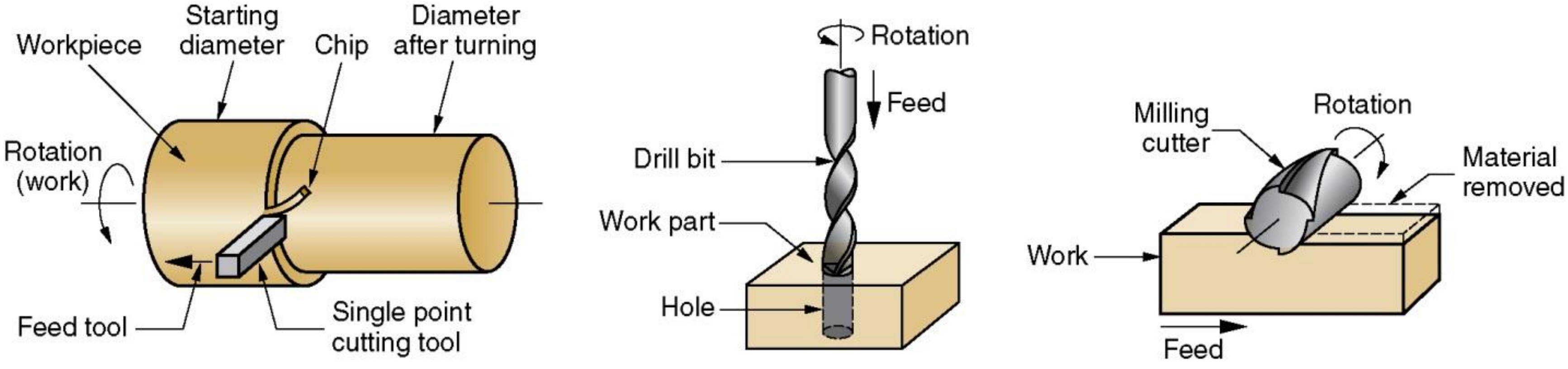
- Forging:
  - https://www.youtube.com/watch?v=EKsiT_-swBM
  - https://www.youtube.com/watch?v=jF7G0feKMSA
  - https://www.youtube.com/watch?v=nBspuKqG9Sw
- Extrusion:
  - https://www.youtube.com/watch?v=vHkwq_2yY9E
  - https://www.youtube.com/watch?v=Y75IQksBb0M

Deformation Processes

- Sheet metal processes:
  - https://www.youtube.com/watch?v=uz1G3z_puug
  - https://www.youtube.com/watch?v=Rn31IEOKgQ8

Material Removal Processes

- Excess material removed from the starting piece so what remains is the desired geometry
  - Examples: (a) turning, (b) drilling, and (c) milling



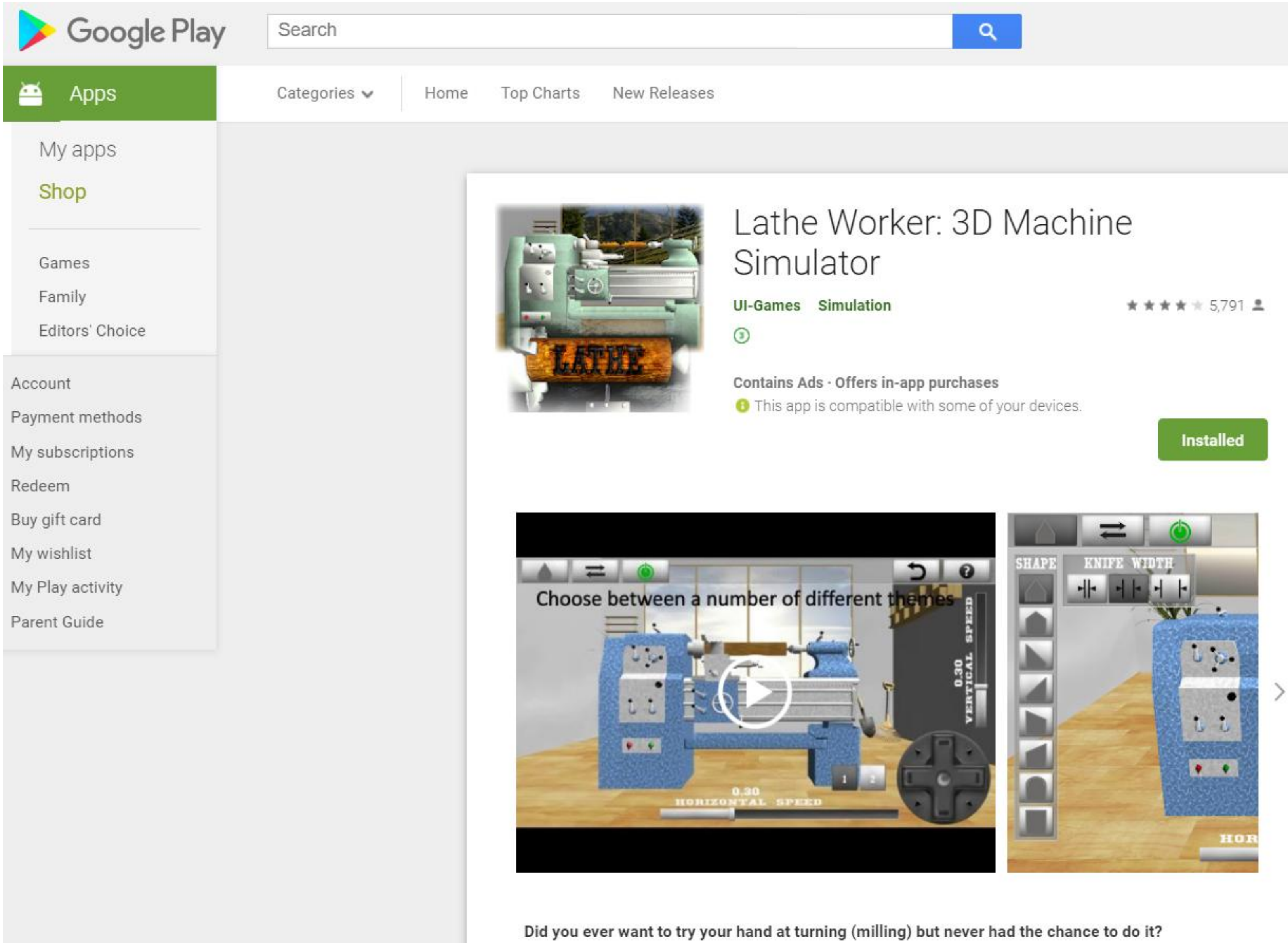
(a) (b) (c)

Material Removal Processes

- Turning:
  - https://www.youtube.com/watch?v=8EsAxOnzEms
- Drilling
- Milling:
  - https://www.youtube.com/watch?v=dkVs5C1HxSo
  - https://www.youtube.com/watch?v=v2wsZI-jsbo

Material Removal Processes

- Gear manufacturing:
  - https://www.youtube.com/watch?v=J1UOjBLo-2M
  - https://www.youtube.com/watch?v=K785jfrsKQU
  - https://www.tec-science.com/mechanical-power-transmission/involute-gear/gear-cutting/
- Broaching:
  - https://www.youtube.com/watch?v=fPkc7PHLIMM



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Buy gift card
My wishlist
My Play activity
Parent Guide
LATHE
Lathe Worker: 3D Machine Simulator
UI-Games
Simulation
5,791
Contains Ads · Offers in-app purchases
This app is compatible with some of your devices.
Installed
Choose between a number of different themes
0.30 VERTICAL SPEED
0.30 HORIZONTAL SPEED
SHAPE
KNIFE WIDTH
Did you ever want to try your hand at turning (milling) but never had the chance to do it?

Waste in Shaping Processes

- It is desirable to minimize waste in part shaping
- Material removal processes are wasteful in the unit operations, but molding and particulate processing operations waste little material
- Terminology for minimum waste processes:
  - *Net shape* processes - little or no waste of the starting material and no machining is required
  - *Near net shape* processes - when minimum machining is required

Property-Enhancing Processes

- Processes that improve mechanical or physical properties of work material
- Examples:
  - Heat treatment of metals and glasses
  - Sintering of powdered metals and ceramics
- Part shape is not altered, except unintentionally
  - Example: unintentional warping of a heat treated part

Surface Processing Operations

- Cleaning - chemical and mechanical processes to remove dirt, oil, and other surface contaminants
- Surface treatments - mechanical working such as sand blasting, and physical processes like diffusion
- Coating and thin film deposition - coating exterior surface of the workpart
- Examples:
  - Electroplating
  - Painting

Assembly Operations

- Two or more separate parts are joined to form a new entity
- Types of assembly operations:
  1. Joining processes – create a permanent joint
     - Welding, brazing, soldering, adhesive bonding
     - https://www.youtube.com/watch?v=GER8xXvFgNI
  2. Mechanical assembly – fastening by mechanical methods
     - Threaded fasteners (screws, bolts and nuts); press fitting, expansion fits
     - https://www.youtube.com/watch?v=kB6ta-IvFxI

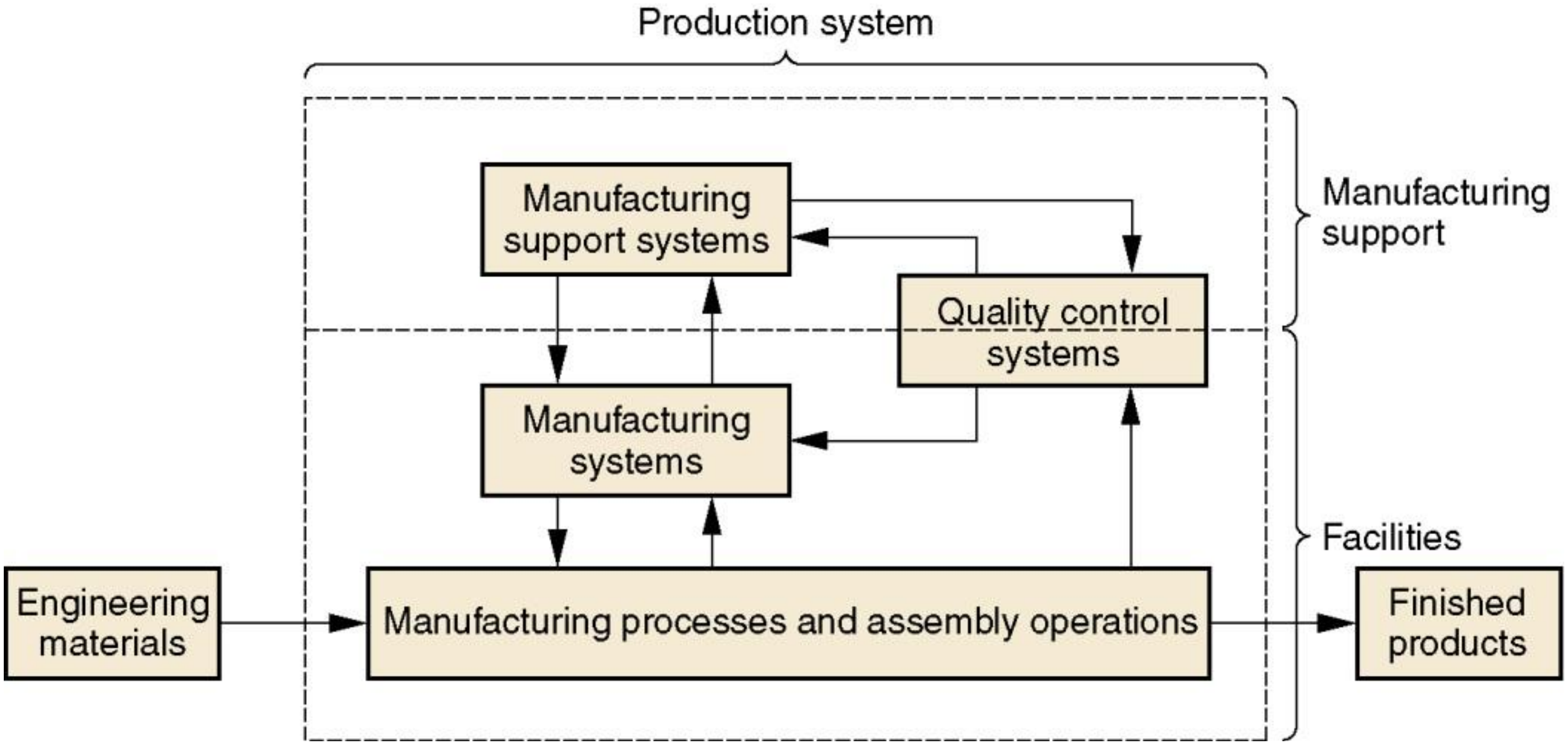
Production Machines and Tooling

- Manufacturing operations are accomplished using machinery and tooling (and people)
- Types of production machines:
  - *Machine tools* - power-driven machines used to operate cutting tools previously operated manually
  - Other production equipment:
    - Presses
    - Forge hammers,
    - Plastic injection molding machines

Production Systems

- People, equipment, and procedures used for the materials and processes that constitute a firm's manufacturing operations
- A manufacturing firm must have systems and procedures to efficiently accomplish its production
- Two categories of production systems:
  - Production facilities
  - Manufacturing support systems
- People make the systems work

Model of the Production System



Production Facilities

- The factory, production equipment, and material handling systems
- Includes the plant layout
- Equipment usually organized into logical groupings, called manufacturing systems
- Examples:
  - Automated production line
  - Machine cell consisting of three machine tools
- Production facilities "touch" the product

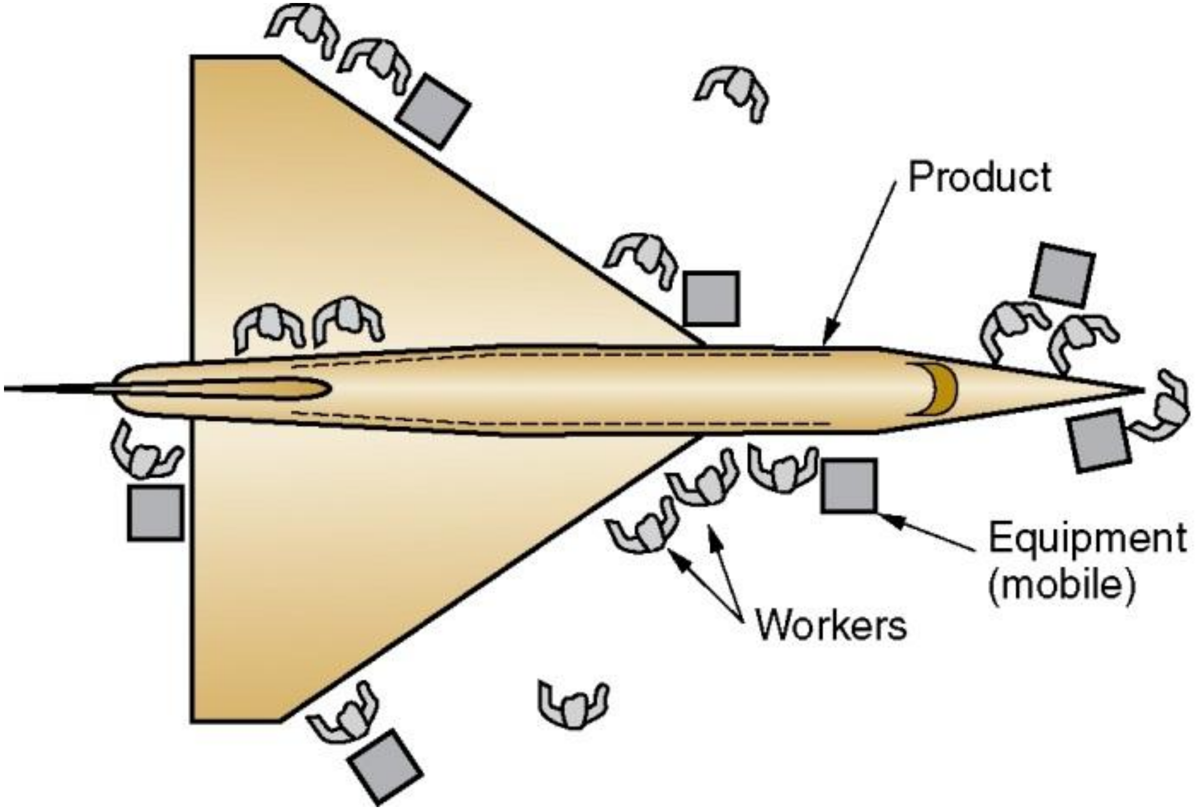
Facilities vs Product Quantities

- A company designs its manufacturing systems and organizes its factories to serve the particular mission of each plant
- Certain types of production facilities are recognized as most appropriate for a given type of manufacturing:
  1. Low production – 1 to 100
  2. Medium production – 100 to 10,000
  3. High production – 10,000 to >1,000,000

Low Production

- *Job shop* is the term used for this type of production facility
- A job shop makes low quantities of specialized and customized products
  - Products are typically complex, e.g., space capsules, prototype aircraft, special machinery
  - Equipment in a job shop is general purpose
  - Labor force is highly skilled
  - Designed for maximum flexibility

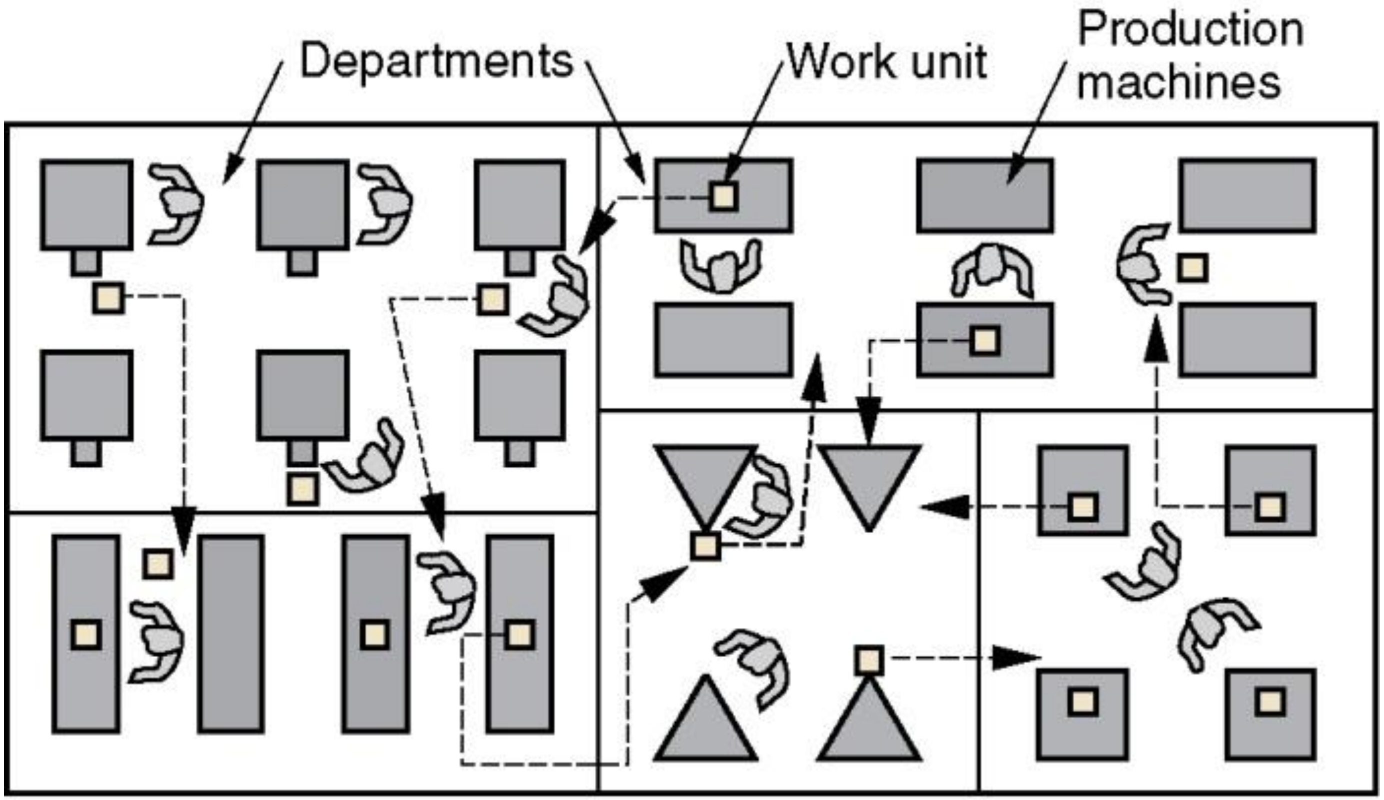
Fixed-Position Plant Layout



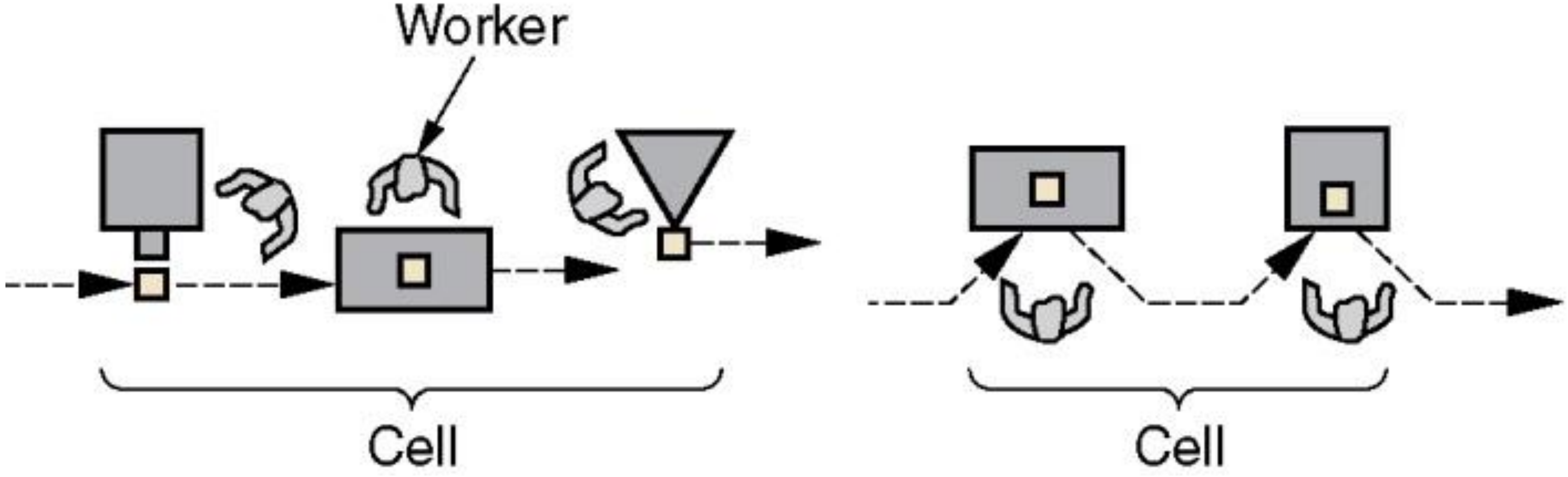
Medium Production

- Two different types of facility, depending on product variety:
  - Batch production
    - Suited to medium and hard product variety
    - Setups required between batches
  - Cellular manufacturing
    - Suited to soft product variety
    - Worker cells organized to process parts without setups between different part styles

Process Plant Layout



Cellular Plant Layout



High Production

- Often referred to as *mass production*
  - High demand for product
  - Manufacturing system dedicated to the production of that product
- Two categories of mass production:
  1. Quantity production
  2. Flow line production

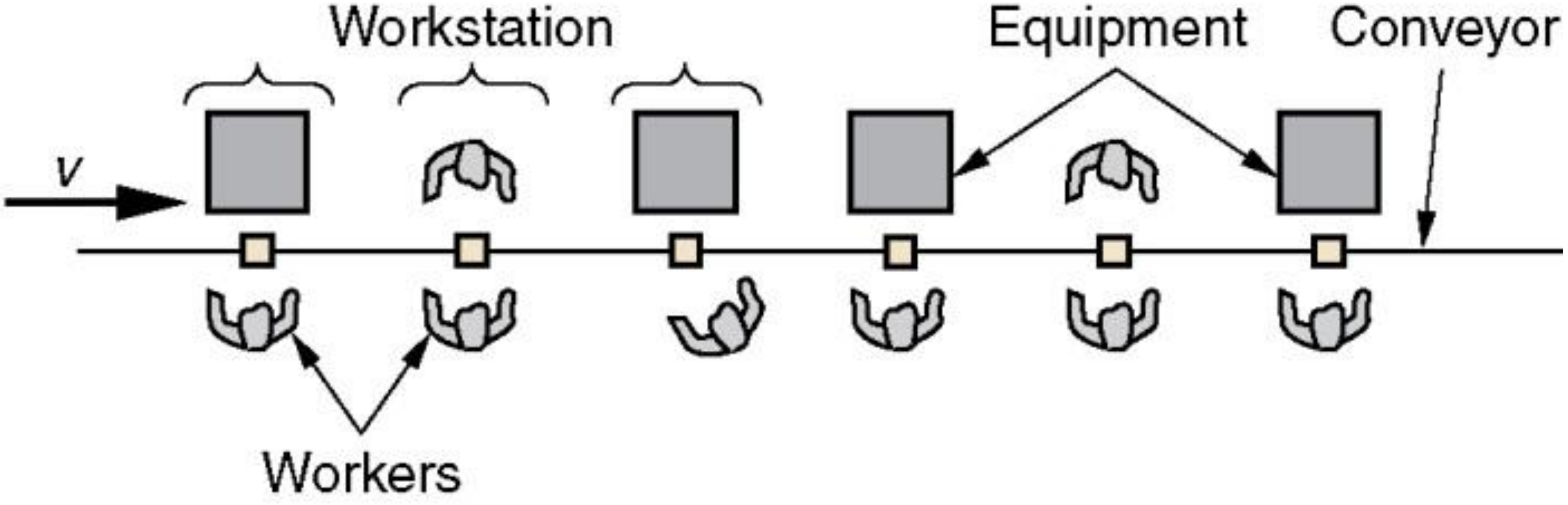
Quantity Production

- Mass production of single parts on single machine or small numbers of machines
  - Typically involves standard machines equipped with special tooling
  - Equipment is dedicated full-time to the production of one part or product type
  - Typical layouts used in quantity production are process layout and cellular layout

Flow Line Production

- Multiple machines or workstations arranged in sequence, as in a production line
  - Product is complex - requires multiple processing and/or assembly operations
  - Work units are physically moved through the sequence to complete the product
  - Workstations and equipment are designed specifically for the product to maximize efficiency

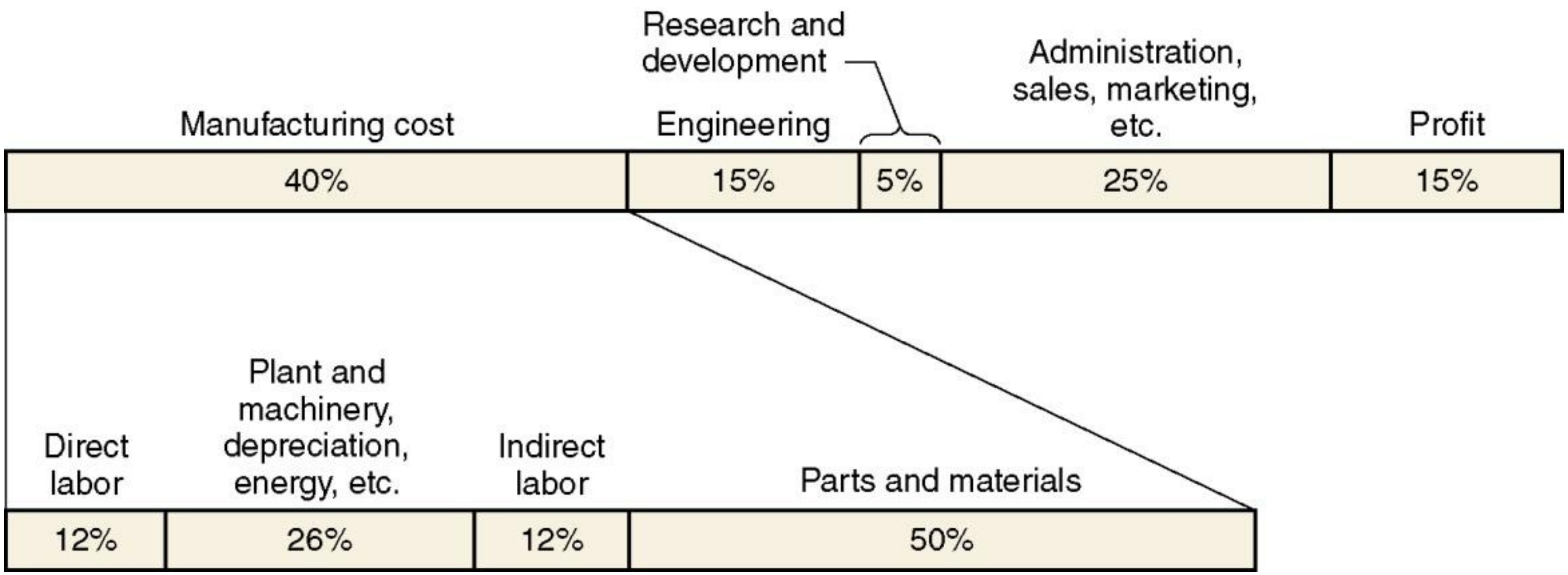
Product Plant Layout



Manufacturing Support Systems

- A company must organize itself to design the processes and equipment, plan and control production, and satisfy product quality requirements
  - Accomplished by *manufacturing support systems*
    - The people and procedures by which a company manages its production operations
  - Typical departments:
    - Manufacturing engineering, Production planning and control, Quality control

Typical Cost Breakdown for a Manufactured Product



Recent Developments in Manufacturing

- Microelectronics
- Computerization in manufacturing
- Flexible manufacturing
- Microfabrication and Nanotechnology
- Lean production and Six Sigma
- Globalization and outsourcing
- Environmentally conscious manufacturing

Microelectronics

- Electronic devices that are fabricated on a microscopic scale: Integrated circuits (ICs)
- Today's fabrication technologies permit billions of components to be included in a single IC
- A large proportion of the products manufactured today are based on microelectronics technology
    - About 2/3 of the products in Table 1.1 are either electronics products or their function and operation depend on electronics

Computerization of Manufacturing

- Direct Numerical Control (DNC) was one of the first applications of computers in manufacturing (1960s)
  - Mainframe computer remotely controlling multiple machine tools
- Enabled by advances in microelectronics, the cost of computers and data processing has been reduced, leading to the widespread use of personal computers
  - To control individual production machines
  - To manage the entire enterprise

Flexible Manufacturing

- Although mass production is widely used throughout the world, computerization has enabled the development of manufacturing systems that can cope with product variety
- Examples:
  - Cellular manufacturing
  - Mixed-model assembly lines
  - Flexible manufacturing systems

Microfabrication and Nanotechnology

- Microfabrication
  - Processes that make parts and products whose feature sizes are in the micron range (10^{-6} m)
  - Examples: Ink-jet printing heads, compact disks, microsensors used in automobiles
- Nanotechnology
  - Materials and products whose feature sizes are in the nanometer range (10^{-9} m)
  - Examples: Coatings for catalytic converters, flat screen TV monitors

Lean Production and Six Sigma

- Lean production
  - Doing more work with fewer resources, yet achieving higher quality in the final product
  - Underlying objective: elimination of waste in manufacturing
- Six Sigma
  - Quality-focused program that utilizes worker teams to accomplish projects aimed at improving an organization's organizational performance

Globalization

- The recognition that we have an international economy in which barriers once established by national boundaries have been reduced
  - This has enabled the freer flow of goods and services, capital, technology, and people among regions and countries
  - Once underdeveloped countries such as China, India, and Mexico have now developed their manufacturing infrastructures and technologies to become important producers in the global economy

Outsourcing

- Use of outside contractors to perform work that was traditionally accomplished in-house
- Local outsourcing
  - Jobs remain in the U.S.
- Outsourcing to foreign countries
  - Offshore outsourcing - production in China and other overseas locations
  - Near-shore outsourcing - production in Canada, Mexico, and Central America

Environmentally Conscious Manufacturing

- Determining the most efficient use of materials and natural resources in production, and minimizing the negative consequences on the environment
- Associated terms: green manufacturing, cleaner production, sustainable manufacturing
- Basic approaches:
  1. Design products that minimize environmental impact
  2. Design processes that are environmentally friendly